higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(Second Paper)
NQF LEVEL 4

7 NOVEMBER 2019

SYMBOL	EXPLANATION
M	Method
A	Accuracy
MA	Method with consistent accuracy
RT/RM	Reading from a table/map
U	Unit
SF	Substitution in formula
MF	Manipulating formula
C	Conversion
R/J	Reasoning/Justification
R	Rounding

This marking guideline consists of 7 pages.

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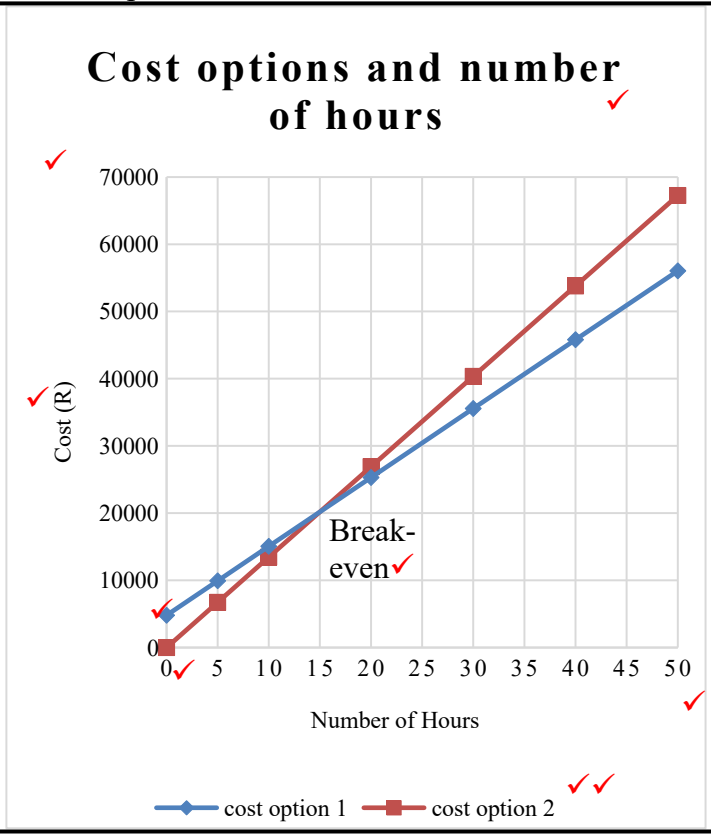
QUESTION 1

Question		Solution	Explanation		
1.1	1.1.1	A = 74 100 + 2 470✓ = 76 570✓ B = 20 020 + 130✓ = 20 150✓ C = (-1560) + (-2470) + (-6500) + (-1560) + 130 + 910✓ = -11 050✓ **OR** C = 650 – 11 700✓ = -11 050✓ **OR** C = (126 750 + 74 100 + 29 250 + 30 290 + 20 150 + 44 070) – (128 310 + 76 570 + 35 750 + 31 850 + 20 020 + 43 160) = 324 610 – 335 660✓ = -11 050✓	1M 1A 1M 1A 1M 1A 1M 1A 1M 1A		(6)
	1.1.2	The flight school made more profit than they had anticipated.	2A		(2)
	1.1.3	$\frac{11\,700}{429\,000} \times 100$ = 2,73 %✓	1RT; 1M 1A		(3)
	1.1.4	To ensure that you are sticking to your budget✓✓ **OR** To enable you to see where you are not staying on budget✓✓ **OR** To help make the next month's budget more accurate✓✓ (Any 1 × 2)	2A		(2)

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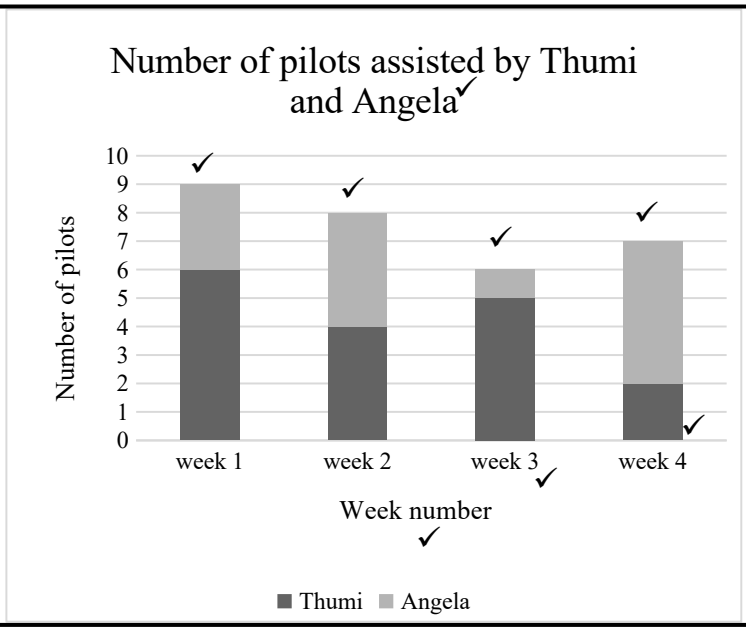
Question	Solution	Explanation		
1.2	Simple: $50\,000 \times \frac{8{,}21}{100}$✓ $= 4\,105$ $4\,105 \times 2$✓ $= 8\,210$ $50\,000 + 8\,210$✓ $=$ R58 210✓ Compound: $50\,000 \times \frac{8{,}13}{100} = 4\,065$✓ $50\,000 + 4\,065 = 54\,065$✓ $54\,065 \times \frac{8{,}13}{100} = 4\,395{,}48$✓ $54\,065 + 4\,395{,}48$✓ $=$R58 460,48✓ Option 2 is the better option.✓	1MA 1MA 1MA 1A 1MA 1MA 1MA 1MA 1A 2R/J		(10)
1.3	Metered option: $6 \times 8{,}28 = 49{,}68$✓ $4 \times 8{,}79 = 35{,}16$✓ $2 \times 15 = 30$✓ $49{,}68 + 35{,}16 + 30 = 114{,}84$✓ $114{,}84 + (114{,}84 \times \frac{15}{100})$✓ $=$ R132,07✓ Prepaid option: $10 \times 7{,}57 = 75{,}70$✓ $2 \times 9{,}72 = 19{,}44$✓ $75{,}70 + 19{,}44 = 95{,}14$✓ $95{,}14 + (95{,}14 \times \frac{15}{100})$ $=$ R109,41✓ The prepaid option is cheaper.✓	1MA 1MA 1MA 1MA 1MA 1A 1MA 1MA 1MA 1A 1R/J		(11)
1.4	$35\,253 + 26\%$ of income above 195 850✓ $= 35\,253 + 26\%$ of $(216\,000 - 195\,850)$✓ $= 35\,253 + (\frac{26}{100} \times 20\,150)$✓ $= 35\,253 + 5\,239$ $= 40\,492$✓ $40\,492 - 14\,067$✓ $= 26\,425$✓ $26\,425 \div 12$✓ $=$ R2 202,08✓	1RT 1SF 1M 1A 1M 1A 1M 1A		(8) **[42]**

QUESTION 2

Question		Solution	Explanation		
2.1	2.1.1	A = R4 800 ✓ B = R25 300 ✓ C = R13 450 ✓ D = 50 hours ✓ E = R 45 800 ✓ F = R 53 800 ✓	1A 1A 1A 1A 1A 1A		(6)
	2.1.2	Cost = 4 800 + ✓ (1 025 × no. of hours)✓	2A		(2)
	2.1.3	Cost✓ depends on the number of hours.✓	2A		(2)
	2.1.4		1 heading 1 labels both axes 1 scale *x*-axis 1 scale *y*-axis 1 break-even point shown 2 y-intercepts 1 label option 1 1 label option 2 (or 2 legend)		(9)
	2.1.5	15 hours✓✓ R20 175✓✓ (accept R20 000 to R20 200)	2A 2A		(4)
2.2	2.2.1	Inverse **OR** Indirect **OR** Decreasing✓ As the number of hours per week increases,✓ the number of weeks decreases.✓	1A 2R/J		(3)
	2.2.2	A = 2 B = 8 C = 6 D = 1	1A 1A 1A 1A		(4)
	2.2.3	24	2A		(2) **[32]**

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QUESTION 3

Question		Solution	Explanation		
3.1	3.1.1	**Mean** 71 285✓ ÷ 6✓ = 11 881✓ Accept 11880 **Median** 0; 12105; 12105; 13450; 16140; 17485✓ (12 105 + 13 450) ÷ 2✓ =12 778✓ **Mode** 12 105✓	2MA 1A 1 arrange 1M 1A 1A		(7)
	3.1.2	Disagree✓ – The mean is lower than most data points✓ due to zero being an outlier.✓	1A 2R/J		(3)
	3.1.3	No,✓ the mode is the highest number in the set.✓✓	1A; 2R/J		(3)
3.2	3.2.1	Yes,✓ Thumi trained more pilots than Angela.✓	1A; 1R/J		(2)
	3.2.2	$\frac{3}{9} = \frac{1}{3}$	2RT 1S		(3)
	3.2.3	$\frac{1}{13} = 7{,}69\%$	2RT 1A		(3)
	3.2.4	$\frac{6}{30}$ OR $\frac{1}{5}$ OR 20% OR 0,2	2RT		(2)
	3.2.5		1 heading 1 labels both axes 1 category labels 1 legend 4 correct bars		(8)
3.3		100 – (25 + 9 + 7 + 6 + 14)✓ = 39%✓ $126\,750 \times \frac{100}{39}$ = 325 000✓	1M 1A 2MA 1A		(5) **[36]**

QUESTION 4

Question		Solution	Explanation		
4.1	4.1.1	Option 1: A = L × B A = 10,6 × 6,8✓ A = 72,08✓ 72,08 × 4✓ = 288,32✓m^2✓	 1SF 1A 1M 1A; 1U		
		Option 2: A = L × B A = 25,5 × 9✓ A = 229,5✓m^2 He should choose option 2 as it takes up a smaller area.✓	 1SF 1A 1R/J		(8)
	4.1.2	25,5m × 100 = 2 550✓cm 2 550 ÷ 8,5✓ = 300✓ 1 : 300✓	1C 1M 1A 1A		(4)
4.2	4.2.1	567,5 ℓ × 1 000✓ = 567 500✓ cm^3✓ V = L × B × H 567 500 = 152,5 × 61 × H✓ H = 567 500 ÷ (152,5 × 61)✓ H = 61,01✓ H ≈ 61 cm✓	1C 1A; 1U 1SF 1MF 1A 1R		(7)
	4.2.2	Radius = 1,54 ÷ 2 = 0,77✓ Volume = π × r^2 × h = 3,14 × $0,77^2$ × 2,048✓ = 3,81 m^3✓ 3,81 × 1 000✓ = 3 810 ℓ✓ 3 810 ÷ 567,5✓ = 6,71✓ The manufacturer's claim is valid.✓	 1A 1SF 1A 1C 1A 1M 1A 1R/J		(8)
4.3	4.3.1	50 mins ÷ 60✓ = 0,83 hours✓ $speed = \frac{distance}{time}$ $speed = \frac{340}{0,83}$ = 409,64✓ km/h	1C 1A 2SF 1A		(5)
	4.3.2	10 ÷ 60 = 0,17✓ 1 + 0,17 = 1,167✓ hours or 70/60✓ = 1,167 hours ✓ 40 ÷ 1,167✓ = 34,28✓ Minimum 35✓ trips	1C 1MA 1M 1A 1R		(5)

	4.3.3	7	1RM		(1)
	4.3.4	Phalaborwa Hoedspruit	2RM		(2) **[40]**

	TOTAL:		**150**

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